

LINUX SERVER ADMINISTRATION & WEB SERVICES LAB

Final Project Report

Operating System

CentOS 7

Server Hostname

linuxsrv

Server IP Address

192.168.10.2

Web Server

Apache HTTP Server

Virtual Hosts

site1.local, site2.local

Remote Administration

SSH

Project Status

Completed Successfully

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1. Project Title

Linux Server Administration & Web Services Lab

2. Project Overview

This project focuses on the practical implementation and administration of a Linux-based server environment using CentOS 7.

The project was designed to develop practical skills in Linux server administration, system configuration, user and group management, file permissions, remote administration, security configuration, Apache web server deployment, virtual hosting, system administration, and troubleshooting.

The Linux server was configured with a static IP address of 192.168.10.2 and used as a centralized server for hosting web services and performing various system administration tasks.

The project also included the deployment of multiple Apache virtual hosts to demonstrate how a single Linux server can host multiple websites.

Throughout the project, configuration, testing, troubleshooting, and verification activities were performed to ensure that all services operated correctly.

3. Project Objectives

The main objectives of this project were to:

- Install and configure a CentOS Linux server.
- Configure the server hostname and static IP address.
- Configure repositories for package management.
- Create and manage Linux users and groups.
- Configure password policies and user privileges.
- Manage Linux files and directories.
- Configure file ownership and permissions.
- Implement Access Control Lists (ACLs).
- Install and configure SSH for remote administration.
- Configure firewall rules using firewalld.
- Understand and configure SELinux security settings.
- Install and configure the Apache HTTP Server.
- Deploy websites on the Linux server.
- Configure Apache Virtual Hosts.
- Host multiple websites on a single server.
- Manage Linux processes and system services.
- Monitor disk usage and filesystem information.
- Analyze system and Apache logs.

- Perform connectivity and web service testing.
- Simulate and troubleshoot an Apache service failure.
- Recover the failed service and verify successful operation.

4. Project Environment

Component	Configuration
Operating System	CentOS 7
Server Hostname	linuxsrv
Server IP Address	192.168.10.2
Server Role	Linux Server / Web Server
Web Server	Apache HTTP Server
HTTP Port	80
Remote Administration	SSH
Firewall	firewalld
Security	SELinux
Virtual Hosts	site1.local, site2.local
Virtualization Platform	VMware / Virtual Machine Environment

5. Network Configuration

The Linux server was configured with a static IP address to ensure consistent network communication and reliable access to the server.

The server configuration used in the project was:

IP Address:	192.168.10.2
Hostname:	linuxsrv

The static IP configuration allowed the server to provide reliable access to the Apache web services and enabled remote administration through SSH.

Network connectivity was tested using standard Linux networking commands such as:

```
ip addr
ip route
ping
```

The server's connectivity to the local network and gateway was successfully verified.

6. Linux Server Setup

The CentOS 7 operating system was installed in a virtual machine environment.

After installation, the basic server configuration was completed.

The following tasks were performed:

- CentOS installation
- Hostname configuration
- Static IP configuration
- Repository configuration
- Network connectivity testing

The hostname was configured as:

```
linuxsrv
```

The server was assigned the static IP address:

```
192.168.10.2
```

The configuration was verified to ensure that the server could communicate correctly with the network.

7. User and Group Administration

Linux users and groups were configured to demonstrate centralized account management on the server.

The following administrative tasks were performed:

- User creation
- Group creation
- Group membership management
- Password management
- User privilege configuration
- Sudo access configuration

User and group management was performed using Linux commands such as:

```
useradd  
passwd  
groupadd  
usermod
```

Administrative privileges were managed using sudo permissions.

This section demonstrated the ability to manage user accounts securely and assign appropriate privileges based on administrative requirements.

8. File System and Permissions

Linux file and directory management was performed to demonstrate access control and data protection.

The following tasks were completed:

- Directory creation
- File creation
- File permissions
- Ownership management
- Group ownership
- Access Control Lists (ACLs)

Linux permissions were configured using:

```
chmod  
chown  
chgrp
```

ACLs were configured using:

```
setfacl  
getfacl
```

The permission system was tested to verify that users had appropriate access to files and directories.

This demonstrated practical understanding of Linux access control and file security.

9. SSH Remote Administration

SSH was installed and configured to provide secure remote administration of the Linux server.

The SSH service was verified using:

```
sudo systemctl status sshd
```

Remote access was tested from a client machine using:

```
ssh <username>@192.168.10.2
```

The successful SSH connection confirmed that the server could be securely administered remotely.

SSH troubleshooting was also performed as part of the project to verify remote connectivity and service availability.

10. Firewall and SELinux Configuration

Linux server security was configured using firewalld and SELinux.

Firewall configuration was used to control network access to server services.

The required services and ports were reviewed and configured according to project requirements.

The Apache HTTP service required access to:

```
TCP Port 80
```

Firewall configuration was tested to ensure that required services were accessible while unnecessary access was restricted.

SELinux was also studied and configured as part of the server security activities.

The project demonstrated the importance of Linux security mechanisms in protecting server services and controlling access.

11. Apache Web Server

The Apache HTTP Server was installed and configured on the CentOS server.

The Apache service was managed using:

```
sudo systemctl start httpd
sudo systemctl stop httpd
sudo systemctl restart httpd
sudo systemctl status httpd
```

The Apache configuration was tested using:

```
sudo apachectl configtest
```

A successful configuration returned:

```
Syntax OK
```

The Apache web server was configured to listen for HTTP requests on port:

```
80
```

The service was tested to verify that web pages could be accessed successfully from client systems.

12. Website Deployment

Web content was deployed on the Linux server using Apache.

Separate document root directories were created for the hosted websites.

The project used:

```
/var/www/site1  
/var/www/site2
```

The websites were tested using a web browser and command-line tools.

The web server successfully delivered the hosted website content to client systems.

13. Apache Virtual Hosting

Virtual hosting was implemented to demonstrate how multiple websites can be hosted on a single Apache server.

Two virtual hosts were configured:

```
site1.local  
site2.local
```

Each website was configured with its own:

- Server name
- Document root
- Access configuration
- Error log
- Access log

The virtual hosts were tested using:

```
curl -H "Host: site1.local" http://192.168.10.2
```

and:

```
curl -H "Host: site2.local" http://192.168.10.2
```

Both virtual hosts successfully returned their respective website content.

This demonstrated practical knowledge of Apache Virtual Host configuration.

14. System Administration

System administration activities were performed to monitor and manage the Linux server.

The following areas were covered:

Process Management

Running processes were viewed using:

```
ps -ef
```

Real-time process monitoring was performed using:

```
top
```

Apache processes were also identified and monitored.

Systemd Service Management

The Apache service was managed using:

```
systemctl
```

The service was stopped, started, and verified.

Apache was also configured to start automatically during system boot.

Disk Management

Disk and filesystem information was checked using:

```
df -h  
df -Th  
lsblk  
fdisk -l
```

These commands were used to monitor available storage and identify disk and filesystem information.

Log Management

System logs were reviewed using:

```
journalctl
```

Apache logs were reviewed from:

```
/var/log/httpd/
```

Access logs and error logs were analyzed to understand server activity and identify potential issues.

15. Testing and Verification

Testing was performed throughout the project to verify that the server and its services were operating correctly.

The following tests were completed:

- IP configuration testing
- Localhost connectivity testing
- Server IP testing
- Default gateway testing
- SSH remote access testing
- Apache service status verification
- HTTP port 80 verification
- Site 1 testing
- Site 2 testing
- Virtual Host testing
- Apache configuration testing
- Log verification

The Apache configuration was verified using:

```
sudo apachectl configtest
```

The HTTP service was verified using:

```
sudo ss -tulpn | grep :80
```

The websites were tested using:

```
curl -H "Host: site1.local" http://192.168.10.2
```

and:

```
curl -H "Host: site2.local" http://192.168.10.2
```

All required services were successfully tested and verified.

16. Troubleshooting Scenario

A controlled Apache service failure was intentionally simulated as part of the troubleshooting exercise.

The Apache service was stopped using:

```
sudo systemctl stop httpd
```

After stopping the service, the hosted websites became inaccessible.

The troubleshooting process followed these steps:

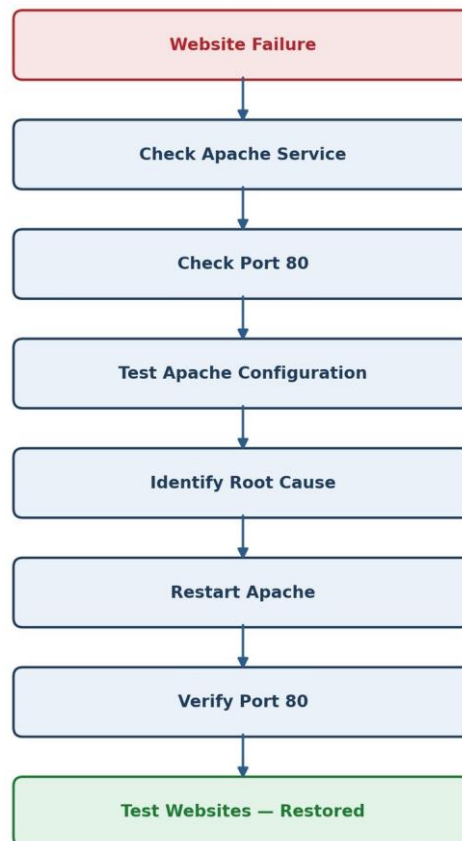


Figure 2: Apache Service Troubleshooting Workflow

The Apache service status was checked using:

```
sudo systemctl status httpd
```

The HTTP port was checked using:

```
sudo ss -tulpn | grep :80
```

The Apache configuration was tested using:

```
sudo apachectl configtest
```

The configuration returned:

```
Syntax OK
```

This confirmed that the Apache configuration was valid.

The root cause was identified as the Apache service being stopped.

The service was restored using:

```
sudo systemctl start httpd
```

After recovery, the Apache service and websites were tested again.

Both websites were successfully restored.

A detailed troubleshooting record is available in:

```
10-Testing-Troubleshooting/04-Troubleshooting-Evidence/Troubleshooting-Report.md
```

17. Problems Encountered and Solutions

Problem 1: Apache Virtual Host Configuration Error

An Apache configuration error was encountered while configuring the virtual hosts.

The error indicated that a closing:

```
</VirtualHost>
```

did not have a matching opening VirtualHost section.

Solution:

The Virtual Host configuration files were reviewed and corrected.

The Apache configuration was then tested using:

```
sudo apachectl configtest
```

The final result was:

```
Syntax OK
```

Problem 2: Apache Service Failure

During the troubleshooting exercise, the Apache service was intentionally stopped.

As a result, the hosted websites became inaccessible.

Solution:

The Apache service status was checked and the service was restarted:

```
sudo systemctl start httpd
```

The service was verified as running and the websites were successfully restored.

18. Project Results

The project successfully achieved its intended objectives.

The final Linux server environment included:

- Static IP: 192.168.10.2
- User & Group Management
- File Permissions & ACLs
- SSH Remote Administration
- Firewall Configuration
- SELinux Security
- Apache Web Server
- Virtual Host 1 — site1.local
- Virtual Host 2 — site2.local
- Process Management
- Systemd Administration
- Disk Management
- Log Management
- Troubleshooting & Recovery

The server was successfully configured, tested, and validated.

19. Skills Demonstrated

This project demonstrated practical skills in:

- CentOS Linux administration
- Linux server installation
- Static IP configuration
- Hostname configuration
- Repository management
- User management
- Group management
- Password management
- Sudo privilege management
- Linux file permissions
- File ownership
- ACL configuration
- SSH administration
- Firewall configuration
- SELinux administration
- Apache web server administration
- Website deployment
- Apache Virtual Hosting
- Process management
- Systemd service management
- Disk management
- Log management
- Network connectivity testing
- Web service testing
- Troubleshooting
- Root cause analysis
- Service recovery

20. Project Evidence

The project evidence was organized into separate directories covering each stage of the implementation.

The evidence includes:

02-Linux-Server-Setup

03-User-Group-Administration

```

04-File-System-Permissions
05-SSH-Remote-Administration
06-Firewall-SELinux
07-Apache-Web-Server
08-Virtual-Hosting
09-System-Administration
10-Testing-Troubleshooting

```

The troubleshooting documentation is available at:

```

10-Testing-Troubleshooting/04-Troubleshooting-Evidence/Troubleshooting-
Report.md

```

Screenshots collected during the project provide evidence of server configuration, service management, testing, and successful operation.

 Add supporting screenshots for each project evidence directory listed above.

21. Conclusion

The Linux Server Administration & Web Services Lab was successfully completed.

The project provided practical experience in deploying and managing a CentOS Linux server in a virtualized environment. The server was configured with a static IP address and prepared for system administration and web hosting tasks.

User and group management, file permissions, ACLs, SSH, firewall configuration, SELinux, Apache web server deployment, virtual hosting, process management, systemd administration, disk monitoring, and log analysis were successfully implemented and tested.

Two Apache virtual hosts were configured and successfully accessed through the Linux server. Testing was performed using both command-line tools and client-side web browsers.

A controlled Apache service failure was also simulated to demonstrate practical troubleshooting skills. The problem was identified, the root cause was determined, the service was restored, and the final system was successfully verified.

Overall, this project demonstrates practical knowledge of Linux server administration, web server management, system monitoring, security configuration, service troubleshooting, and operational support.

The project provides a strong foundation for further work in Linux System Administration, Network Administration, NOC Operations, and IT Infrastructure Management.

22. Final Project Status

Project Status: COMPLETED SUCCESSFULLY

Task	Status
------	--------

Linux Server Setup	✓ Completed
User & Group Administration	✓ Completed
File System Permissions	✓ Completed
SSH Administration	✓ Completed
Firewall & SELinux	✓ Completed
Apache Web Server	✓ Completed
Virtual Hosting	✓ Completed
System Administration	✓ Completed
Testing & Troubleshooting	✓ Completed
Final Verification	✓ Successful

Final Result: Linux Server Administration & Web Services Lab Successfully Completed